

WASP-104b

R-2521

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_180317 · created 2026-08-25T01:20:07+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458194.8727 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.044
Transit depth (Rp/Rs)^2	2.2 +/- 1.29 [%]
Orbital Inclination (inc)	83.36 +/- 2.84
Airmass coefficient 1 (a1)	1.01 +/- 0.026
Airmass coefficient 2 (a2)	-0.005 +/- 0.02
Scatter in the residuals of the lightcurve fit is	2.58 %
Best Comparison Star	#2 - [332, 318]
Optimal Aperture	2.92
Optimal Annulus	6.79
Transit Duration (day)	0.066 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.044 vs 0.1214 (deviation 0.6σ)
Duration fit vs published	0.066 d vs 0.0742 d (deviation 0.27σ)
Mid-time O–C	3.4 min
Depth SNR	3.4
Residual scatter	2.58 %

Light curve

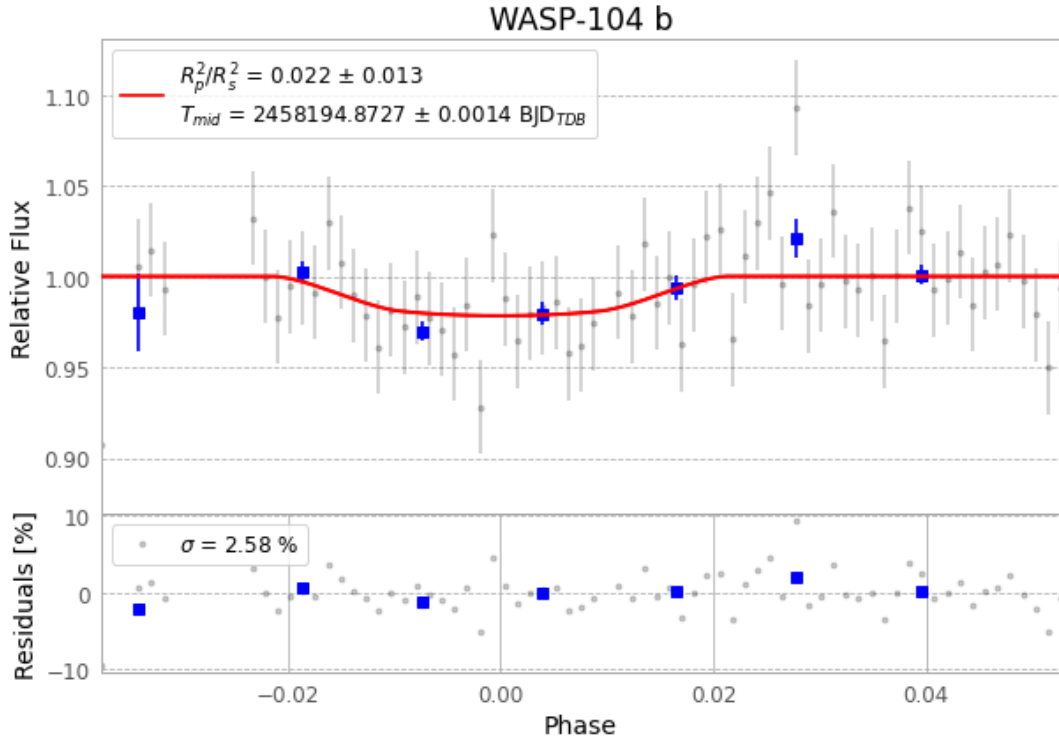


Figure 1 — FinalLightCurve_WASP-104 b_2018-03-17.png (SHA-256 380283d28ce57da5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9e03c16f0e475e45...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

77 FITS frames (51 MB), WASP-104180317071212.FITS → WASP-104180317110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-180317.log (e827d0dceab6...), FinalLightCurve_WASP-104 b_2018-03-17.png (380283d28ce5...), FinalLightCurve_WASP-104 b_2018-03-17.csv (2a12cd8aa593...)

Compute resources

Wall time 436.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 01:20Z.